

HHP – HULDRA HALEPRODUKSJON

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SYSTEM CONTROL DIAGRAM - HVAC SYSTEM, HCM SEQUENCES

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START HVAC, HCM SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait before next step	Jump condition	Jump to step	Remarks
Start permissives	0	NA	NA	1 or 3	System not available: 77-GK71A.XE or 77-GK71B.XE or 77-GK102A.XE or 77-GK102B.XE = TRUE	System available: 77-GK71A.XE and 77-GK71B.XE and 77-GK102A.XE and 77-GK102B.XE = FALSE	3	
Abort because equipment unavailable	1	1: Warning on OS	1: 77-XA9101.X	END	NA	NA	NA	
Open all dampers Indication on OS: "Åpne spjeld"	3	1: Open all dampers 2: Start timer 45 sec	1: 77-GS71.XH 2: 77-GS72.XH 3: 77-GS73.XH 4: 77-GS74.XH (One signal to F03)	4 or 5	Timer expired	All dampers not closed: 77-ZSL2013 and 77-ZSL2014 and 77-ZSL2016 and 77-ZSL2017 = FALSE	5	
Abort because of damper malfunction	4	1: Start HVAC Stop sequence 2: Warning on OS	1: 77-HS9100L.LA 2: 77-HS9100L.XH 3: 77-XA9102.X	END	NA	NA	NA	
Start A fans	5	1-2: Start A fans 3: Start fans timeout timer 45 sec	1: 77-GK71A.XP1H 2: 77-GK102A.XF1H 3: 77-GK71A.LA 4: 77-GK102A.LA	6 or 7	Timer expired	A fans running: 77-GK71A.BCH and 77-GK102A.BCH = TRUE	7	
Abort because A fans malfunction	6	1: Start HVAC Stop sequence 2: Warning on OS	1: 77-HS9100L.LA 2: 77-HS9100L.XH 3: 77-XA9103.X	END	NA	NA	NA	
Start B fans	7	1-2: Start B fans 3: Start fans timeout timer 45 sec	1: 77-GK71B.XP1H 2: 77-GK102B.XP1H 3: 77-GK71B.LA 4: 77-GK102B.LA (One signal from F03)	8 or 9	Timer expired	B fans running: 77-GK71B.BCH and 77-GK102B.BCH = TRUE	9	
Abort because B fans malfunction	8	1: Start HVAC Stop sequence 2: Warning on OS HCM fans not running	1: 77-HS9100L.LA 2: 77-HS9100L.XH 3: 77-XA9104.X	END	NA	NA	NA	
Ten seconds delay Indication on OS: "Pause i 10 sekunder"	9	1: Start timer 10 sec		10	Timer expired	NA	NA	
Switch on heater: Indication on OS: "Slå på varme"	10	1: Start heater 2: Start heater timeout timer 8 sec	1: 77-FE101.XP1H 2: 77-FE101.LA	11 or END	Timer expired	Heater on: 77-FE101.BCH = TRUE	END	
Alarm because of heater mafunction	11	Warning on OS		END	NA	NA	NA	
END	END	NA	NA	NA	NA	NA	NA	

System Control Diagram

HVAC, HCM Sequences

STOP HVAC, HCM SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait before next step	Jump condition	Jump to step	Remarks
Start permissives	0	Set all fans and heater to auto	1: 77-GK71A.LA 2: 77-GK71B.LA 3: 77-GK102A.LA 4: 77-GK102B.LA 5: 77-FE101.LA	1	NA	NA	NA	
Switch off heater Indication on OS "Slå av varme"	1	1: Stop command to heater	1: 77-FE101.XP1L	2	Heater confirmed off: 77-FE101.BCL = TRUE	NA	NA	
Three minutes delay Indication on OS: "Pause i 3 minutter"	2	1: Start timer 180 sec		3	Timer expired	NA	NA	
Stop A fans Indication on OS "Stopp A vifter"	3	1-2: Stop A fans	1: 77-GK71A.XP1L 2: 77-GK102A.XP1L	4	Fans confirmed stopped: 77-GK71A.BCL and 77-GK102A.BCL = TRUE	NA	NA	
Stop B fans Indication on OS "Stopp B vifter"		1-2: Stop B fans	1: 77-GK71B.XP1L 2: 77-GK102B.XP1L	5	Fans confirmed stopped: 77-GK71B.BCL and 77-GK102B.BCL = TRUE	NA	NA	
Close all dampers Indication on OS: "Lukke spjeld"	4	1: Close all dampers	1: 77-GS71.XL 2: 77-GS72.XL 3: 77-GS73.XL 4: 77-GS74.XL (One signal to F03.)	END	All dampers closed: 77-ZSL2016 and 77-ZSL2017 and 77-ZSL2013 and 77-ZSL2014 = TRUE	NA	NA	
END	END	NA	NA	NA	NA	NA	NA	